Agentic Automation
(with Managerial Accounting)

Acctg & Fin | BUS AN 500 | Class 05
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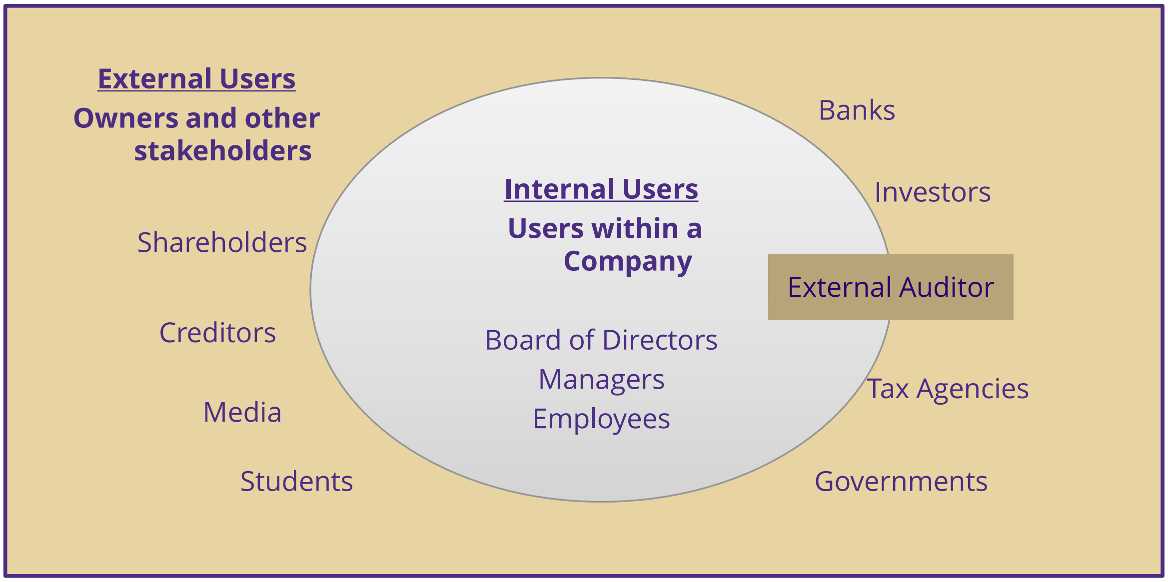
What will we do today?

- **Review**
  - **Accounting & finance function**
  - **Intelligent automation demo**
- Managerial accounting overview
  - Budgeting and control
- Variance Analysis Lab
  - Understanding the manual process
  - RPA Steps v. agents
  - Humans: Policy decision making
- Time permitting, final project in-class time

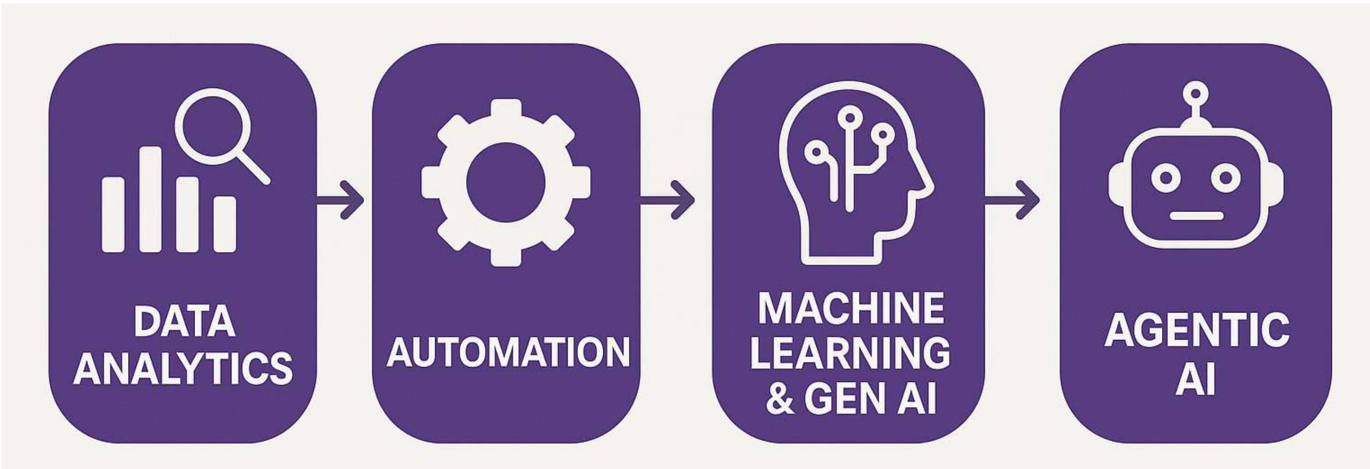
Accounting & Finance Function



An information and decisions approach

	Financial Accounting	Managerial Accounting
Users and decision makers	External: Investors, creditors, and other external users	Internal: Managers, employees
Purpose of the information	Help external users make investment, credit and other decisions	Help managers make planning and control decisions
Flexibility of reporting	Structure, follows GAAP requirements plus voluntary disclosures	Flexible; can be structured as to best inform managers
Timeliness of information	Public after period end after an audit	Available on demand
Time dimension	Historical information with some projections	Future information, including projections and forecasts backed with historical data
Focus of information	Aggregate entity-level with some segmented data	Detailed around projects, processes and divisions
Nature of information	Monetary with voluntary non-monetary	Monetary and non-monetary including KPIs

Process Diagram



How can we use data analytics and automation tools to improve decisions?

Financial reporting example

Task:

From an external user's perspective, understand Apple, Inc's revenue growth and drivers of that growth.

Why:

We want to consider an investment in APPL.

Financial reporting e.g., two manual steps

Task:

From an external user's perspective, understand Apple, Inc's revenue growth and drivers of that growth.

Manual approach:

- Calculate revenue growth / trends by collecting revenue data for APPL.
- Collect information about how APPL talks about it's revenue (e.g., in the Management Discussion and Analysis section of the 10-K)

Financial reporting e.g., manual step 1

Task:

From an external user's perspective, understand Apple, Inc's revenue growth and drivers of that growth.

Manual approach:

- Calculate revenue growth / trends by collecting revenue data for APPL.
    - Maybe use the 10-K? Collect from the income statement?
    - Use a third party, like Yahoo! Finance?
    - Enter into spreadsheet and calculate year-over-year growth.

Financial reporting e.g., manual step 2

Task:

From an external user's perspective, understand Apple, Inc's revenue growth and drivers of that growth.

Manual approach:

- Collect information about how APPL talks about it's revenue (e.g., in the Management Discussion and Analysis section of the 10-K)
    - Get the 10-K, find the MD&A in the index and go to that section.
    - Read and make notes from the discussion.
    - Make a judgement about the discussion that supports revenue growth.

What can RPA do here?

RPA approach:

✓ Calculate revenue growth / trends by collecting revenue data for APPL.

  - Connect to SEC's API and download AAPL's disclosed Revenues, calculate the year-over-year change, report back as a Table.

✓ Collect information about how APPL talks about it's revenue (e.g., in the Management Discussion and Analysis section of the 10-K)

  - Download the 10-K filing, use a rule to identify the MD&A section, return it as text (removing html tags).
  - ~~Read and make notes from the discussion.~~
  - ~~Make a judgement about the discussion that supports revenue growth.~~

What can an Intelligent RPA workflow add?

RPA passes:

- ✓ Revenue growth table and MD&A section to the LLM based agent.
- ✓ LLM agent can then:
  - Summarize the revenue trends and find textual associations linked to revenue growth (like us reading and making notes).
  - Provide insight to support a judgement about the discussion that supports revenue growth by providing a requested output.

Intelligent Automation Demo

Financial Reporting: AAPL example

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What is managerial accounting

An internal (to the firm) user's perspective on using accounting information to support:

- Planning, directing and motivating employees, (cost) controlling, and evaluation responsibilities of managers.

Why is it important?

Managerial accounting tools:

- Transforms raw data into actionable decisions
- Identifies where managerial attention is needed
- Distinguishes signal from noise in performance (and performance evaluation)
- Enables timely intervention (not just post-hoc analysis)
- Serves as a trigger for action (not just measurement)

Managerial Decisions

Managerial decisions that can be supported by managerial accounting tools include examples like:

- Should we rent or buy a new location?
  - Can be very important for retail companies (e.g., Starbucks)
- Should we onboard or outsource an activity?
  - Can be very important for service firms (e.g., CPAs and banks)
- Should we build an internal AI data warehouse or rely on OpenAI / Claude subscriptions?
  - Can be very important for software companies (e.g., SAAS)

What is common between these decisions?

In each case, accounting information helps identify **when a decision is required what trade-offs matter:**

- Should we rent or buy a new location?
  - **Trade-off between fixed commitments and flexibility**
- Should we onboard or outsource an activity?
  - **Trade-off between control, cost, and scalability**
- Should we build an internal AI data warehouse or rely on OpenAI / Claude subscriptions?
  - **Trade-off between upfront investment and usage-based costs**

The commonality:

These are all **trade-off decisions about trade-offs when managers face uncertainty using (historical and projected) cost information.**

These decisions are often triggered by:

- cost changes
- performance gaps
- or **unexpected variances**

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Variance Analysis Lab Overview

Using Variance Analysis as a setting, we will think about the following decision process:

- **Variance analysis** → "Where did expectations break?"
- **Automation (RPA)** → "We can detect this automatically"
- **Agent (LLM)** → "Here's what matters and what to do"
- **Human** → "Should we act, and how?"

Our lab question

Our firm is (almost) 10% over budget. Is this a problem, or just noise? What should we do?

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The manual process, create report

To manually (or automatically) create a report we need:

- Budget data (usually aggregated to a unit / project level)
- Actual spending data (usually not aggregated)

Steps:

1. Obtain both datasets
2. Aggregate actuals to same units as budgets
3. Join the budget and actual data

budget

File Edit View Insert Format Data Tools Extensions Help

A1 BudgetID

	A	B	C	D	E	F
1	BudgetID	Period	Department	Account	Category	BudgetAmount
2	B-2026-04-001	2026-04	Sales	Travel	Operating Expense	8500
3	B-2026-04-002	2026-04	Sales	Meals & Entertainment	Operating Expense	2200
4	B-2026-04-003	2026-04	Marketing	Digital Ads	Operating Expense	18000
5	B-2026-04-004	2026-04	Marketing	Events	Operating Expense	12000
6	B-2026-04-005	2026-04	Operations	Shipping	COGS / Fulfillment	15000
7	B-2026-04-006	2026-04	Operations	Supplies	Operating Expense	6500
8	B-2026-04-007	2026-04	Finance	Software	Operating Expense	4200
9	B-2026-04-008	2026-04	Finance	Professional Fees	Operating Expense	9500
10	B-2026-04-009	2026-04	HR	Training	Operating Expense	5000
11	B-2026-04-010	2026-04	IT	Cloud Services	Operating Expense	7800
12	B-2026-04-011	2026-04	IT	Hardware	Capital / Equipment	6000
13	B-2026-04-012	2026-04	Admin	Office Rent	Fixed Expense	21000

actual_transactions

TransactionID	Date	Period	Department	Account	Vendor	Description	Amount	Approver	PaymentMetho	ReceiptAttached
A-10001	4/2/2026	2026-04	Sales	Travel	Delta Airlines	Customer visit airfare	2450.75	Sales Director	Corporate Car	Yes
A-10002	4/4/2026	2026-04	Sales	Travel	Marriott	Customer visit hotel	1795.2	Sales Director	Corporate Car	Yes
A-10003	4/9/2026	2026-04	Sales	Travel	Uber	Local transportation	628.4	Sales Director	Corporate Car	Yes
A-10004	4/18/2026	2026-04	Sales	Travel	Alaska Airlines	Late-booked customer visit	4850	Sales Director	Corporate Car	Yes
A-10005	4/11/2026	2026-04	Sales	Meals & Entert	Local Bistro	Client dinner	780.1	Sales Director	Corporate Car	Yes
A-10006	4/20/2026	2026-04	Sales	Meals & Entert	Convention Ce	Client event meal package	1935	Sales Director	Corporate Car	No
A-10007	4/3/2026	2026-04	Marketing	Digital Ads	Google Ads	Paid search campaign	9400	Marketing Man	ACH	Yes
A-10008	4/15/2026	2026-04	Marketing	Digital Ads	Meta	Social media campaign	7600	Marketing Man	ACH	Yes
A-10009	4/25/2026	2026-04	Marketing	Digital Ads	LinkedIn	Executive audience campaign	3950	Marketing Man	ACH	Yes
A-10010	4/5/2026	2026-04	Marketing	Events	NW Trade Sho	Booth registration	6500	Marketing Man	ACH	Yes
A-10011	4/7/2026	2026-04	Marketing	Events	DisplayCo	Booth materials	3400	Marketing Man	Credit Card	Yes
A-10012	4/26/2026	2026-04	Marketing	Events	FreightExpress	Rush shipping for event materials	2750	Marketing Man	Credit Card	No
A-10013	4/2/2026	2026-04	Operations	Shipping	UPS	Outbound shipping	8200	Operations Ma	ACH	Yes
A-10014	4/16/2026	2026-04	Operations	Shipping	FedEx	Outbound shipping	6850	Operations Ma	ACH	Yes
A-10015	4/28/2026	2026-04	Operations	Shipping	DHL	Expedited customer order shipping	3100	Operations Ma	ACH	Yes
A-10016	4/8/2026	2026-04	Operations	Supplies	Packaging Plu	Boxes and labels	3150	Operations Ma	Credit Card	Yes
A-10017	4/22/2026	2026-04	Operations	Supplies	LabelWorks	Replacement printer labels	2480	Operations Ma	Credit Card	Yes
A-10018	4/1/2026	2026-04	Finance	Software	QuickBooks	Accounting subscription	950	Controller	ACH	Yes
A-10019	4/10/2026	2026-04	Finance	Software	DataViz Cloud	Analytics subscription	1650	Controller	ACH	Yes
A-10020	4/21/2026	2026-04	Finance	Software	Automation Cl	New workflow licenses	2250	Controller	ACH	Yes
A-10021	4/12/2026	2026-04	Finance	Professional F	Advisory LLP	Monthly advisory support	9400	Controller	ACH	Yes
A-10022	4/27/2026	2026-04	Finance	Professional F	Advisory LLP	Additional review hours	2100	Controller	ACH	Yes
A-10023	4/14/2026	2026-04	HR	Training	SkillUp	Employee training workshop	4100	HR Manager	Credit Card	Yes
A-10024	4/19/2026	2026-04	HR	Training	SkillUp	Additional seats	1450	HR Manager	Credit Card	Yes
A-10025	4/6/2026	2026-04	IT	Cloud Services	AWS	Cloud hosting	5200	IT Manager	ACH	Yes
A-10026	4/17/2026	2026-04	IT	Cloud Services	Azure	Backup services	1900	IT Manager	ACH	Yes
A-10027	4/29/2026	2026-04	IT	Cloud Services	AWS	Unexpected compute spike	1850	IT Manager	ACH	Yes
A-10028	4/9/2026	2026-04	IT	Hardware	Dell	Replacement laptops	5990	IT Manager	ACH	Yes
A-10029	4/30/2026	2026-04	Admin	Office Rent	Lake Union Pr	Monthly rent	21000	Office Manage	ACH	Yes

The manual process, analyze report

A manager receives a variance report and:

1. Reviews overall budget vs actual
2. Identifies large or unusual variances
3. Drills into departments / accounts / units / projects
4. Investigates potential causes
5. Determines whether the variance is:
    1. signal (requires action)
    2. noise (can be ignored)
6. Decides whether to act
7. Communicates findings / recommendations

variance_analysis

File Edit View Insert Format Data Tools Extensions Help

A1 fx BudgetID

	A	B	C	D	E	F	G	H	I	J
1	BudgetID	Period	Department	Account	Category	BudgetAmount	ActualAmount	Variance (Budge	Variance (%)	Outcome
2	B-2026-04-012	2026-04	Admin	Office Rent	Fixed Expense	21000	21000	0	0	Favorable
3	B-2026-04-008	2026-04	Finance	Professional Fees	Operating Expense	9500	11500	-2000	-17.39130435	Unfavorable
4	B-2026-04-007	2026-04	Finance	Software	Operating Expense	4200	4850	-650	-13.40206186	Unfavorable
5	B-2026-04-009	2026-04	HR	Training	Operating Expense	5000	5550	-550	-9.90990991	Unfavorable
6	B-2026-04-010	2026-04	IT	Cloud Services	Operating Expense	7800	8950	-1150	-12.84916201	Unfavorable
7	B-2026-04-011	2026-04	IT	Hardware	Capital / Equipment	6000	5990	10	0.1669449082	Favorable
8	B-2026-04-003	2026-04	Marketing	Digital Ads	Operating Expense	18000	20950	-2950	-14.08114558	Unfavorable
9	B-2026-04-004	2026-04	Marketing	Events	Operating Expense	12000	12650	-650	-5.138339921	Unfavorable
10	B-2026-04-005	2026-04	Operations	Shipping	COGS / Fulfillment	15000	18150	-3150	-17.3553719	Unfavorable
11	B-2026-04-006	2026-04	Operations	Supplies	Operating Expense	6500	5630	870	15.45293073	Favorable
12	B-2026-04-002	2026-04	Sales	Meals & Entertainment	Operating Expense	2200	2715.1	-515.1	-18.97167692	Unfavorable
13	B-2026-04-001	2026-04	Sales	Travel	Operating Expense	8500	9724.35	-1224.35	-12.59055875	Unfavorable
14						115700	127659.45	-11959.45	-9.368244967	Unfavorable

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Where does the manual process become difficult?

1. Large volume of data / large numbers of (unfavorable) variances
2. Time constraints
3. Inconsistent judgment / Limited visibility into causes
4. If it is hard to prioritize what matters

RPA helpers versus Agent helpers

Step	RPA (Automation)	Agent (Interpretation)
Data ingestion	Reads budget and actuals	—
Aggregation	Computes totals and variances	—
Detection	Flags differences	—
Interpretation	—	Explains what happened
Prioritization	—	Identifies what matters most
Recommendation	—	Suggests actions
Decision	—	Supports human judgment

RPA helpers versus agent helpers

- RPA answers: **“What happened?”**
- Agents answer: **“What does it mean and what could we do?”**
- Humans **decide whether a change is needed**

Lab time

- See and Fill in the rest of the template for today's class:
  1. Use an LLM to get insight from the variance and actuals data.
  2. Add the policy document as a follow-up and ask for recommendations on rewriting the policy.
  3. Reflection activity.

Team check in

Decide on your focus area as a team, internal (manager) or external (investor) decision maker?

- Choose a potential question you'd like to answer within a broad area – (see slides in the appendix) or pick a broad area and I'll supply a question.

Thank you

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Appendix

Reminder of Final Project Requirements

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Intelligent Automation Team Challenge

A team project that focuses on an enterprise level automation robot aimed at solving an Accounting & Finance problem.

- The bot can be demonstrated in a proof-of-concept state.
- Has a Generative Artificial Intelligence (LLM) and/or Agentic AI / AI Agents component.

Requirements

- Software deliverable: A prototype (or example) of a solution to the problem.

- Presentation deliverable: Discussion of the problem being solved, how the agent benefits the task or workflow, and an evaluation of the usefulness of the Agent or LLM example.

Accounting & Finance Focus Areas

We will focus on the following core areas:

- Financial Performance & Forecasting
- Budgeting & Financial Planning and Analysis (FPA)
- Project Evaluation & Risk

Teams have the option to choose from these areas (or some other options).

Structuring the problem

I will provide each team with a problem from within the Accounting & Finance Function* for this problem, you will be asked to present an AI-agent solution to the use-case focusing on:

- What does an intelligent agent need to know, compute, and determine/reason?

* If your team is interested in solving a different problem from the pool of potential ones I offer let me know

Structuring the project

The project will have each team focus on

- What are the appropriate inputs/data?
- What is the model or logic used to solve the problem?
- What decisions and explanations do I want the agent's help on?

Financial Analysis & Forecasting

Data: transactions, financial statements, earnings reports

Logic: Trend analysis & ratio analysis

Agent Role: Explain performance & generate insights, predict outcomes & explain key assumptions

Users/audience: Internal (managers) and external (investors) to the firm

Budgeting and & Financial Planning and Analysis

Data: Transactions / budgets versus actuals reports (internal).

Logic / tools: variance analysis, relevant costs, cost allocation.

Agent roles: Diagnose what went wrong (missing a budget target), aid in filtering decision-relevant information (relevant v irrelevant costs).

Users/audience: Internal management decisions.

Project Evaluation & Risk

Data: investment targets, risk/return estimates and cost of capital, projected future cash-flows.

Logic tools: Time value of money, valuation, capital asset pricing models.

Agent roles: Rank & Recommend Investment decisions based on a set of targets, simulate uncertainty and risk analysis and/or evaluate investment trade-offs over time.

Users/audience: Mangers and investors.

Some other potential focus areas

- Fraud Detection
- Portfolio Management
- Environmental, Social and Governance reporting
- Personal Finance
- Wealth Planning
- Tax Optimization
- Compliance
- Cybersecurity
- Blockchain

Structuring the data

Simulate data internal company inputs for this task or use real company disclosures and/or stock prices.

Proof-of-concept / prototype

Multiple possible approaches:

- Goal: Proof of agent concept (UiPath, Python, LLM) without automation **do this part first.
- Goal: Proof of concept process map in business process mapping notation (BPMN) **do this second.
- Goal: Working automation workflow prototype with agents using business process mapping notation (UiPath, Python). **optional; do this third.

Presentation Preparation

Each team will present for up to 15 minutes plus up to a 5 minute Q&A.

- What is the problem, why is it important
- What your solution achieves
- Mechanics of the solution (data inputs, process, outputs)
- Conclusion and Q&A

Non-presenting class members are expected to ask questions.